Subject: ENVIRONMENTAL IMPACT ASSESSMENT

Subject Code: 23A01504c

UNIT - III

Assessment of Impact on Vegetation, Wildlife and Risk Assessment

1. Introduction

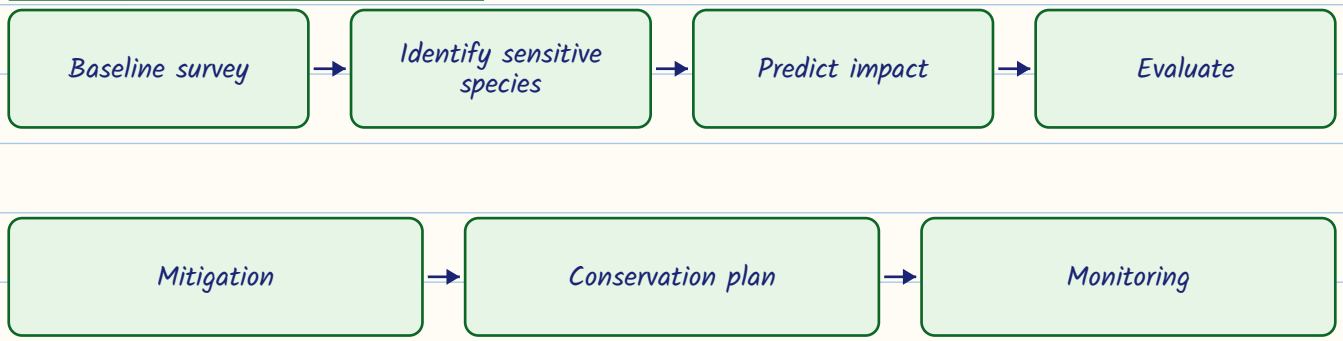
Vegetation (flora) and wildlife (fauna) form the biological environment. They give us food, timber, medicine, clean air, climate regulation and beauty. Developmental projects destroy or disturb them, so their assessment is a very important part of EIA.

- **Biodiversity** the variety of life at gene, species and ecosystem levels.
- India is one of the 17 mega-diverse countries and has 4 biodiversity hotspots (Himalaya, Western Ghats, Indo-Burma, Sundaland - Nicobar).
- Biological components are good indicators of the health of the environment, because plants and animals respond to pollution and habitat change.

Aim of the biological assessment to know what species are present, how sensitive they are, how the project will affect them and how the loss can be avoided, reduced or compensated.

2. Assessment of Impact of Development Activities on Vegetation and Wil

2.1 Steps of the assessment



2.2 Baseline data collection

- **Secondary data** Forest Department records, Botanical Survey of India (BSI), Zoological Survey of India (ZSI), Wildlife Institute of India, satellite images and published literature.
- **Primary data** field surveys in different seasons.
- **Vegetation survey methods** quadrat method (1 m x 1 m for herbs, 5 m x 5 m for shrubs, 10 m x 10 m or 20 m x 20 m for trees), line transect and point-centred quarter method.
- **Wildlife survey methods** line transect and direct sighting, pug marks and scat (dung) counts, camera traps, bird point counts, nets for small animals, interviews with local people.
- **Aquatic life** plankton net, benthic sampler and fish catch.

2.3 Parameters studied

Parameter	Meaning
Density	Number of individuals per unit area
Frequency	Percentage of quadrats in which a species occurs
Abundance	Number of individuals per quadrat of occurrence
Basal area / Biomass	Size and weight of vegetation
Species diversity	Shannon-Wiener index H = - sum of (pi x ln pi)
Importance Value Index (IVI)	Relative density + relative frequency + relative dominance

- Also recorded: endemic, rare, endangered and threatened (RET) species as per IUCN Red List, Schedules of Wild Life Protection Act and BSI/ZSI lists.
- Note nearby protected areas such as national parks, sanctuaries, biosphere reserves, elephant corridors and Ramsar wetlands.

2.4 Types of impact

- **Direct impacts** clearing of vegetation, loss of habitat, killing of animals by vehicles and blasting, drowning by reservoirs.

- **Indirect impacts** air and water pollution reduce growth; dust on leaves reduces photosynthesis; noise and light disturb animals; increased human access leads to poaching and fire.
- **Habitat fragmentation** roads, canals and railway lines divide the forest into small patches; animals cannot move, breed or feed properly; gene flow reduces.
- **Edge effect** change in conditions at the edge of a cut forest.
- **Barrier effect** dams stop fish migration; fences stop animal movement.
- Introduction of invasive alien species (e.g. Lantana, Eichhornia - water hyacinth).
- Cumulative impacts from many projects in the same region.

Activity wise impacts

Activity	Impact on flora and fauna
Dams and reservoirs	Submergence of forest, fish migration blocked, downstream flow change
Mining	Loss of forest, dust, overburden dumps, water pollution
Roads and railways	Fragmentation, road kills, noise
Thermal power plant	Fly ash, SO2 damage to plants, thermal discharge
Industries	Toxic effluents, bioaccumulation in food chain
Urbanisation	Loss of habitat, light and noise pollution

2.5 Prediction and evaluation

- Predict the area of vegetation lost, number of trees to be felled, change in species diversity and loss of habitat for key species.
- **Methods** GIS and remote sensing for land cover change, habitat suitability models, population viability analysis, expert judgement and analogy with similar projects.
- **Significance depends on** rarity of species, ecological value, size of the area lost, whether it can be restored, and legal protection.

- Higher importance is given to schedule I species (tiger, elephant, lion), endemic species, breeding sites and corridors.

2.6 Mitigation measures

- Avoid sensitive areas by changing the site or alignment.
- Minimise clearing; use existing roads and degraded land.
- Compensatory afforestation (twice the degraded land area or equal area of non-forest land as per Forest Conservation Act 1980).
- Green belt of native species around the project.
- Wildlife corridors, underpasses and overpasses on highways, fish ladders at dams.
- Translocation of rare plants and animals; seed banks and nurseries.
- Restrict noise, lights and construction at breeding seasons.
- Strict control on poaching, fire and grazing; awareness for workers and villagers.
- Wildlife conservation plan and monitoring, budget given in the EMP.

3. Environmental Impact of Deforestation

Deforestation means the permanent removal of forest cover and its conversion to other uses such as agriculture, mining, industry, roads or settlement.

- India's forest cover is about 21 percent of the geographical area (national policy target is 33 percent).
- Tropical forests are lost mainly for agriculture, timber and development projects.

3.1 Causes of deforestation

- **Agricultural expansion** shifting (jhum) cultivation and conversion of forest to farm land.
- **Mining and quarrying** open cast mines destroy large areas.

- **River valley projects** dams submerge forests.
- **Industrialisation and urbanisation** factories, townships, roads.
- Fuel wood and fodder collection by rural people.
- Commercial logging for timber, paper and plywood.
- Overgrazing by cattle.
- Forest fires, natural or man made.
- Encroachment and illegal felling.
- **Infrastructure** roads, railways, power lines, defence needs.

3.2 Effects of deforestation

Area	Effect
Soil	Erosion, loss of fertility, landslides, siltation of rivers and reservoirs
Water	Floods in rainy season, drought in dry season, lower groundwater recharge, poor water quality
Climate	Less rainfall, rise in local temperature, more CO2 (global warming), loss of carbon sink
Biodiversity	Loss of habitat, extinction of species, human-animal conflict
Economy	Loss of timber, non timber forest products, medicinal plants, tourism
Society	Displacement and loss of livelihood of tribal people

- Desertification may occur in dry regions.
- Loss of the gene pool that could be useful for medicine and agriculture.
- Rise in air pollution as trees no longer filter dust and gases.

3.3 Control measures

- Afforestation, reforestation and social forestry.
- Strict implementation of Forest Conservation Act and Wild Life Act.
- Use of alternative fuels such as LPG and biogas; fuel efficient stoves.
- Sustainable forest management and Joint Forest Management (JFM) with local communities.

- Controlled grazing, fire prevention and watch towers.
- Use of recycled paper; agro-forestry and wasteland development.
- Remote sensing for monitoring forest cover.

Remember Chipko movement (1973, Uttarakhand, led by Sunderlal Bahuguna and Gaura Devi) is a famous people's movement to save trees.

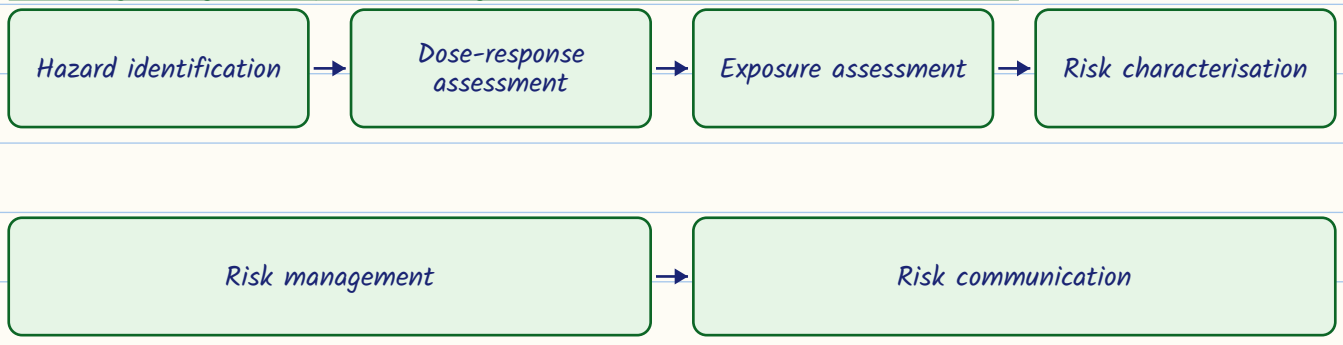
4. Risk Assessment and Treatment of Uncertainty

4.1 Introduction

- **Risk** the chance (probability) that a harmful event will occur, multiplied by how serious its consequence is. Risk = Probability x Consequence.
- Hazard is the source of danger (a toxic chemical, a fire); risk is the chance that the hazard actually harms someone.
- Environmental Risk Assessment (ERA) is the scientific process of estimating the likelihood and the size of adverse effects on human health and ecosystem because of exposure to hazards.
- ERA is used for hazardous industries, chemical storage, waste disposal sites, pipelines and accidents like Bhopal gas tragedy (1984).

Types of risk health risk, ecological risk, accident (safety) risk and natural hazard risk.

4.2 Key stages in performing environmental risk assessment



- **Stage 1 – Hazard identification** find the chemicals, processes and events that

may cause harm and what type of harm (toxic, carcinogenic, fire, explosion). Tools: HAZOP, FMEA, fault tree and event tree analysis, checklists.

- **Stage 2 – Dose-response (toxicity) assessment** relation between the amount of exposure (dose) and the effect on the organism. Uses LD50, LC50, NOAEL and reference dose (RfD) from lab and epidemiological data.
- **Stage 3 – Exposure assessment** who is exposed, by which route (inhalation, ingestion, skin contact), how much, how often and for how long. Uses dispersion models and food chain models.
- **Stage 4 – Risk characterisation** combining the above to give a numerical value of risk, for example excess cancer cases per million people, or hazard quotient (HQ = exposure / reference dose; HQ above 1 means concern).
- **Stage 5 – Risk management** decide what is acceptable and apply controls – engineering controls, safety distances, emergency plans, insurance.
- **Stage 6 – Risk communication** tell the public and authorities about the risk honestly and clearly.

Acceptable risk levels

- A lifetime cancer risk of 1 in a million (10 to the power minus 6) is generally considered negligible; up to 1 in 10,000 may be tolerated in some cases.
- Individual risk and societal risk (F-N curve) are both considered for major hazards.

4.3 Consequence analysis (for accidents)

- **Possible accidents** toxic gas release, fire, pool fire, jet fire, vapour cloud explosion, BLEVE.
- Models predict the distance up to which heat radiation, overpressure or toxic concentration crosses a harmful limit.
- Output is a map of damage zones used for the on-site and off-site emergency plan, siting and buffer zones.

- Disaster Management Plan (DMP) includes alarm system, evacuation, first aid, fire fighting and communication.

4.4 Treatment of uncertainty

Uncertainty means that predictions are not exactly known because of missing data, simple models or natural variation.

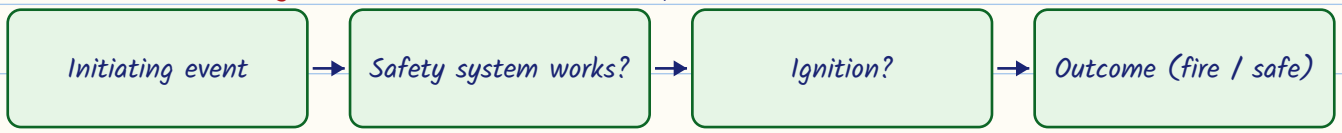
- **Sources** errors in measurement and sampling, variability among people and species, model assumptions, extrapolation from animals to humans or from high dose to low dose, and incomplete knowledge.
- **Types** parameter uncertainty, model uncertainty and scenario uncertainty. Variability (real differences) is different from uncertainty (lack of knowledge).
- **Ways to treat it**
  - Sensitivity analysis - change one input at a time and see its effect on the result.
  - Monte Carlo simulation - run the model thousands of times with random inputs from probability distributions and get a range of results.
  - Use of safety (uncertainty) factors, usually 10 each for animal to human and for human to human variation.
  - Worst case and best case scenarios; conservative assumptions.
  - Confidence intervals and probability statements.
  - Collect more data, peer review and expert opinion.
  - Adaptive management - monitor and adjust after the project starts.

Key idea uncertainty cannot be removed completely, but it should be identified, quantified and clearly shown to decision makers.

4.5 Hazard identification tools in brief

- **HAZOP (Hazard and Operability study)** a team examines each part of a process using guide words such as NO, MORE, LESS, REVERSE, OTHER THAN, and asks what deviation can occur, its causes, consequences and safeguards.

- **FMEA (Failure Mode and Effect Analysis)** lists every component, how it can fail and what effect the failure has.
- **Fault tree analysis (FTA)** a top-down diagram that starts from an undesired top event (for example, tank rupture) and finds all the combinations of causes using AND and OR gates.
- **Event tree analysis (ETA)** a bottom-up diagram that starts from an initiating event (for example, gas leak) and follows the possible outcomes (ignition or no ignition, safety system working or failing).
- **What-if analysis and checklists** simple tools for small units.



4.6 Simple risk numerical

A chemical plant has a probability of tank failure of 0.002 per year. If it happens, the expected number of deaths is 50.

- Risk = Probability x Consequence
- Risk = 0.002 x 50 = 0.1 deaths per year

Result expected loss is 0.1 deaths per year. If the company installs a safety valve that reduces the probability to 0.0004, then the new risk is 0.02 deaths per year, a fall of 80 percent. This shows how risk assessment helps in judging whether a safety investment is worth it.

4.7 Hazard quotient example

- A pollutant has a reference dose RfD = 0.02 mg/kg/day.
- Estimated exposure of people through drinking water = 0.05 mg/kg/day.
- HQ = Exposure / RfD = 0.05 / 0.02 = 2.5
- Since HQ is greater than 1, there is a health concern and the source must be controlled.

4.8 Ecological risk assessment

- It estimates the risk to plants, animals and ecosystems from stressors such as chemicals, habitat loss or invasive species.
- **Steps** problem formulation, analysis (exposure and effects) and risk characterisation.
- Uses indicator species, bioassays (LC50 on fish and Daphnia), bioaccumulation and biomagnification data.
- **Bioaccumulation** build up of a chemical in an organism. Biomagnification :: increase in concentration at each higher level of the food chain (for example DDT and mercury).

5. Advantages of Environmental Risk Assessment

- Gives a scientific and logical basis for decisions.
- Helps in identifying the most serious hazards so that money is spent where it is most useful.
- Allows comparison of different options and alternatives.
- Helps in setting standards and priorities.
- Reduces chances of accidents, health damage and loss of property.
- Helps in preparing emergency and disaster management plans.
- Increases public confidence and transparency through risk communication.
- Helps in legal compliance and in obtaining clearances and insurance.
- Encourages cleaner technology and safer design.

Limitations

- Needs a lot of data and skilled persons.
- Results depend on assumptions and are uncertain.
- Hard to measure ecological and long term effects.
- Public may not understand numerical risk values.

5A. Biodiversity Conservation in Projects

- **In situ conservation** protecting species in their natural habitat - national parks, wildlife sanctuaries, biosphere reserves, sacred groves.
- **Ex situ conservation** outside the habitat - zoos, botanical gardens, gene banks, seed banks, cryopreservation.
- **Protected area categories in India** national parks (no human activity allowed), wildlife sanctuaries (limited activity), conservation reserves, community reserves and biosphere reserves.
- Eco-sensitive zones (ESZ) around protected areas, usually up to 10 km, where industries and mining are regulated.
- **Species protection** IUCN Red List categories are Extinct, Critically Endangered, Endangered, Vulnerable, Near Threatened and Least Concern.
- Project proponent must prepare a Wildlife Conservation Plan with the State Wildlife Warden when a Schedule I species is present within the study area.

Case Example - Silent Valley (Kerala)

- A hydroelectric project proposed on the Kunthipuzha river in the 1970s threatened the evergreen rainforest and the lion tailed macaque.
- Strong protests by scientists and the public stopped the project; the area was declared a National Park in 1984.
- **Lesson** ecological value can be more important than short term power benefits; EIA should include biological studies at the very start.

Case Example - Sardar Sarovar and Tehri dams

- Submergence of forests and agricultural land and displacement of thousands of families.
- Shows the link between biological, social and cultural impacts; resettlement and rehabilitation (R&R) plans must be part of the EIA.

6. Case Example – Bhopal Gas Tragedy (1984)

- Leak of methyl isocyanate (MIC) from the Union Carbide pesticide plant on the night of 2-3 December 1984.
- Thousands of people died and lakhs were affected; long term health and soil and groundwater contamination followed.
- **Causes** storage of large quantity of MIC, poor maintenance, failure of safety systems, plant near a densely populated area, no proper emergency plan.
- **Lessons** need for hazard analysis, safe siting, buffer zones, emergency preparedness and public awareness. Led to the Environment Protection Act 1986 and the Public Liability Insurance Act 1991.

7. Key Points to Remember

- Vegetation is studied by quadrats and transects; fauna by transects, pug marks and camera traps.
- Diversity is measured by the Shannon-Wiener index.
- **Main causes of deforestation** agriculture, mining, dams, industries, fuel wood, logging.
- Risk = probability x consequence.
- **ERA stages** hazard identification, dose-response, exposure assessment, risk characterisation, risk management and communication.
- Uncertainty is treated by sensitivity analysis, Monte Carlo simulation and safety factors.

Important questions (1) Explain the impact of development activities on vegetation and wildlife. (2) Causes and effects of deforestation. (3) Explain the key stages in environmental risk assessment. (4) How is uncertainty treated? (5) Advantages of environmental risk assessment.